SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	Bijoy Varghese	**Reg. No.:**	192411013
Section / Batch:		**Date:**	20/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
		GRAND TOTAL:	**100 Marks**

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I Bijoy Varghese (192411013) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Bijoy Varghese

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)	4	3	4	3	3	43
Q2 (50 Marks)	4	3	3	4	3	42.5
					Overall Total (100)	85

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: ______________	Date: ______________	Date: ______________

Q1) Online Video Streaming Platform.

Recurrence = $T(n) = 2T(n/2) + n$.

$a = 2$

$b = 2$

$f(n) = n$

$n^{\log_b a} = n^{\log_2 2} = n$

Since

$f(n) = \Theta(n^{\log_b a})$

this satisfies Master Theorem Case 2

Therefore:

$T(n) = \Theta(n \log n)$

Analysis:

* Processing time grows as $n \log n$.
* Suitable for handling large no. of users.
* Perform much better than quadratic algorithms.

Final Answer:

Time Complexity : $\Theta(n \log n)$

Q2) E-Commerce Recommendation System.

Algorithms:

Algorithm A = $O(n \log n)$

Algorithm B = $O(n^2)$

Comparison:

Algorithm	Growth Rate	Performance.
$O(n \log n)$	Slow growth	Efficient
$O(n^2)$	Fast growth.	Less Efficient.

For very large datasets;

$$n \log n < n^2$$

Hence, Algorithm A perform much faster.

Conclusion:

* Best Algorithm: $O(n \log n)$
* Better scalability.
* Lower execution time.

Q3) Final Financial Analytics System.

Recurrence:

$T(n) = 2T(n/2) + n \log n$

Given,

$a = 2, b = 2, f(n) = n \log n$

Compute:

$n^{\log_b a} = n$

Since,

$f(n) = \Theta(n \log n) = \Theta(n^{\log_b a} \log^1 n)$

Apply Master Theorem Case 2 $(K = 1)$

Therefore:

$T(n) = \Theta(n \log^2 n)$

Analysis

* Slightly slower than $n \log n$
* Efficient for large financial dataset
* Suitable for stock market analysis.

Final Answer.

✓ Time Complexity : $\Theta(n \log^2 n)$

Q4) Banking Transcation Processing System.

$T(n) = 4T(n/2) + n^2$

Given:

$a = 4$

$b = 2$

$f(n) = n^2$

Compute

$n^{\log_2 4} = n^2$

Since:

$f(n) = \Theta(n^2)$

Master Theorem Case 2 Applies

Hence:

$T(n) = \Theta(n^2 \log n)$

Analysis:

* Processing increases as $n^2 \log n$
* Suitable only for moderate transaction volumes.

Final Answer:

Time Complexity : $\Theta(n^2 \log n)$

Q5) Search Engine Index Builder.

Recurrence:

$$T(n) = T(n/2) + 1.$$

Substitution Method.

$$T(n) = T(n/2) + 1$$
$$= T(n/4) + 2$$
$$= T(n/8) + 3$$

Continue untill

$$n/2^k = 1$$

Therefore:

$$k = \log_2 n.$$

Hence

$$T(n) = T(1) + \log n$$

Ignoring Constant

$$T(n) = \Theta(\log n)$$

Analysis:

* Recursion depth is $\log n$
* Very efficiency for large dataset.

Final Answer:

Time Complexity $= \Theta(\log n)$